

LKB ENGINEERING PRIVATE LIMITED
E-418 TO E-421 , CHOPANKI INDUSTRIAL AREA , BHIWADI ,
DISTT. ALWAR , RAJASTHAN-301019

(AN ISO CERTIFIED , NABL ACCREDITED AND BIS CERTIFIED LABORATORY)

TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3.5C X 300 SQMM	3.5C X 240 SQMM
1.0	General			
a	Brand Name	---	RALLISON	RALLISON
b	Standards applicable	---	IS: 17505 (Part 1) :2021 with up to date amendment	IS: 17505 (Part 1) :2021 with up to date amendment
c	Voltage Grade	Volts	1100 Volts	1100 Volts
d	Permissible voltage & frequency variation	%	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$
e	Suitable for earthed / unearthed systems		Suitable for earth and un earthed system	Suitable for earth and un earthed system
2.0	Current rating			
a	Continuous current rating under normal conditions			
i	In Ground at 30°C	Amps	469	418
ii	In Air at 40°C	Amps	574	500
b	Maximum conductor temperature for normal operation	°C	90	90
c	Maximum conductor temperature for short circuit rating	°C	250	250
d	Short circuit rating for 1 sec. duration	K.Amps	42.93	34.344
3.0	Conductor	---		
a	Material	---	Annealed Plain Copper	Annealed Plain Copper
b	Standard applicable	---	as per IS : 8130-2013 , Class-2	as per IS : 8130-2013 , Class-2
c	Shape of Conductor	---	Stranded, Compacted & Sector Shaped	Stranded, Compacted & Sector Shaped
d	No. of strands	Nos.	37/19	37/19
e	Dia of Strands (Before stranding)	mm	Diameter of strands shall be so choosen to meet the conductor Resistance	Diameter of strands shall be so choosen to meet the conductor Resistance
f	Nominal Cross Sectional Area Phase	sqmm	300	240
g	Nominal Cross Sectional Area Neutral	sqmm	150	120
h	Max. Conductor Resistance Phase at 20°C(DC)	Ω /KM	0.0601	0.075
i	Max. Conductor Resistance Neutral at 20°C(DC)	Ω /KM	0.124	0.153
j	Max. Conductor Resistance Phase at 90°C	Ω /KM	0.076928	0.096512
k	Approximate Reactance of cable at 50 Hz for ref only	Ω /KM	0.072	0.073
l	Approximate Capacitance at 50Hz for ref. only	μ F/Km	0.33	0.31
4.0	Fire Barrier Tape		Glass Mica Tape (Fire Barrier Tape) 2 Tape shall be applied by helical wrapping with 100 % Coverage ,Over conductor throught the length .	

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CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3.5C X 300 SQMM	3.5C X 240 SQMM
5.0	Insulation	---		
a	Material	---	XLPE	XLPE
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Nominal Thickness (Phase)	mm	1.8	1.7
d	Nominal Thickness (Neutral)		1.4	1.2
e	Minimum volume resistivity at 27°C	Ω-cm	1X10 ¹⁴	1X10 ¹⁴
f	Core identification		Red, Yellow, Blue & Black	Red, Yellow, Blue & Black
6.0	Inner Sheath	---		
a	Material	---	Thermoplastic material (PVC with FR Properties)	Thermoplastic material (PVC with FR Properties)
b	Nominal Thickness	mm	1.6	1.6
c	Minimum Thickness	mm	1.08	1.08
d	Extruded/ Wrapped		Extruded	Extruded
e	Colour		Black	Black
7.0	Armouring	---		
a	Material	---	Galvanized Steel	Galvanized Steel
b	Standard applicable	---	IS:3975/1999	IS:3975/1999
c	Type of Armour		Round Wire	Round Wire
d	Size of armour	mm	3.15 ± 0.080	2.5 ± 0.065
e	Binder Tape Over Armour		Synthetic tape may be applied by helical wrapping with 100 % coverage over armour	
8.0	Outer Sheath	---		
a	Material	---	LSZH	LSZH
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Minimum Thickness	mm	2.52	2.2
d	Nominal Thickness	mm	3.4	3.0
e	Overall diameter	mm	68	61
f	Tolerance in overall diameter	± mm	± 4	± 4
g	Outer Sheath colour		Black	Black
8.1	Circuit Integrity Test		Shall pass the test Categories C as per IS:17505 (Part-1) :2021 at 950 ±40 0 C for 3 Hours Category F	
9.0	Min. Bending Radius		12 X Cable Dia	12 X Cable Dia
10.0	Packing Detail			
a	Type of Packing Drum		Wooden Drum	Wooden Drum
b	Packing Length with Tolerance	metres	250 ± 5%	250 ± 5%
c	Order Quantity Tolerance		± 5%	± 5%
11.0	Marking on Outer Sheath			
a	By Printing		RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No	RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No
b	Sequential Length Marking		Shall be provided at every metere by printing	Shall be provided at every metere by printing

Note: 5 % of Ordered quantity may be supplied in Assorted Non-Standard length but shall not be less than 50 Metres

Note: In case of any dispute/typographical error interpretation of the IS specification shall prevail.

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CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3.5C X 185 SQMM	3.5C X 150 SQMM
1.0	General			
a	Brand Name	---	RALLISON	RALLISON
b	Standards applicable	---	IS: 17505 (Part 1) :2021 with up to date amendment	IS: 17505 (Part 1) :2021 with up to date amendment
c	Voltage Grade	Volts	1100 Volts	1100 Volts
d	Permissible voltage & frequency variation	%	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$
e	Suitable for earthed / unearthed systems		Suitable for earth and un earthed system	Suitable for earth and un earthed system
2.0	Current rating			
a	Continuous current rating under normal conditions			
i	In Ground at 30°C	Amps	362	321
ii	In Air at 40°C	Amps	422	365
b	Maximum conductor temperature for normal operation	°C	90	90
c	Maximum conductor temperature for short circuit rating	°C	250	250
d	Short circuit rating for 1 sec. duration	K.Amps	26.4735	21.465
3.0	Conductor	---		
a	Material	---	Annealed Plain Copper	Annealed Plain Copper
b	Standard applicable	---	as per IS : 8130-2013 , Class-2	as per IS : 8130-2013 , Class-2
c	Shape of Conductor	---	Stranded, Compacted & Sector Shaped	Stranded, Compacted & Sector Shaped
d	No. of strands	Nos.	37/19	19/19
e	Dia of Strands (Before stranding)	mm	Diameter of strands shall be so choosen to meet the conductor Resistance	Diameter of strands shall be so choosen to meet the conductor Resistance
f	Nominal Cross Sectional Area Phase	sqmm	185	150
g	Nominal Cross Sectional Area Neutral	sqmm	95	70
h	Max. Conductor Resistance Phase at 20°C(DC)	Ω /KM	0.0991	0.124
i	Max. Conductor Resistance Neutral at 20°C(DC)	Ω /KM	0.193	0.268
j	Max. Conductor Resistance Phase at 90°C	Ω /KM	0.126848	0.15872
k	Approximate Reactance of cable at 50 Hz for ref only	Ω /KM	0.073	0.073
l	Approximate Capacitance at 50Hz for ref. only	μ F/Km	0.29	0.29
4.0	Fire Barrier Tape		Glass Mica Tape (Fire Barrier Tape) 2 Tape shall be applied by helical wrapping with 100 % Coverage ,Over conductor throught the length .	

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CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3.5C X 185 SQMM	3.5C X 150 SQMM
5.0	Insulation	---		
a	Material	---	XLPE	XLPE
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Nominal Thickness (Phase)	mm	1.6	1.4
d	Nominal Thickness (Neutral)		1.1	1.1
e	Minimum volume resistivity at 27°C	Ω-cm	1X10 ¹⁴	1X10 ¹⁴
f	Core identification		Red, Yellow, Blue & Black	Red, Yellow, Blue & Black
6.0	Inner Sheath	---		
a	Material	---	Thermoplastic material (PVC with FR Properties)	Thermoplastic material (PVC with FR Properties)
b	Nominal Thickness	mm	1.4	1.4
c	Minimum Thickness	mm	0.92	0.92
d	Extruded/ Wrapped		Extruded	Extruded
e	Colour		Black	Black
7.0	Armouring	---		
a	Material	---	Galvanized Steel	Galvanized Steel
b	Standard applicable	---	IS:3975/1999	IS:3975/1999
c	Type of Armour		Round Wire	Round Wire
d	Size of armour	mm	2.50 ± 0.065	2.0 ± 0.050
e	Binder Tape Over Armour		Synthetic tape may be applied by helical wrapping with 100 % coverage over armour	
8.0	Outer Sheath	---		
a	Material	---	LSZH	LSZH
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Minimum Thickness	mm	2.2	1.88
d	Nominal Thickness	mm	3.0	2.6
e	Overall diameter	mm	55	49
f	Tolerance in overall diameter	± mm	± 4	± 4
g	Outer Sheath colour		Black	Black
8.1	Circuit Integrity Test		Shall pass the test Categories C as per IS:17505 (Part-1) :2021 at 950 ±40 0 C for 3 Hours Category F	
9.0	Min. Bending Radius		12 X Cable Dia	12 X Cable Dia
10.0	Packing Detail			
a	Type of Packing Drum		Wooden Drum	Wooden Drum
b	Packing Length with Tolerance	metres	250 ± 5%	500 ± 5%
c	Order Quantity Tolerance		± 5%	± 5%
11.0	Marking on Outer Sheath			
a	By Printing		RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No	RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No
b	Sequential Length Marking		Shall be provided at every metere by printing	Shall be provided at every metere by printing

Note: 5 % of Ordered quantity may be supplied in Assorted Non-Standard length but shall not be less than 50 Metres

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CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3.5C X 120 SQMM	3.5C X 95 SQMM
1.0	General			
a	Brand Name	---	RALLISON	RALLISON
b	Standards applicable	---	IS: 17505 (Part 1) :2021 with up to date amendment	IS: 17505 (Part 1) :2021 with up to date amendment
c	Voltage Grade	Volts	1100 Volts	1100 Volts
d	Permissible voltage & frequency variation	%	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$
e	Suitable for earthed / unearthed systems		Suitable for earth and un earthed system	Suitable for earth and un earthed system
2.0	Current rating			
a	Continuous current rating under normal conditions			
i	In Ground at 30°C	Amps	287	254
ii	In Air at 40°C	Amps	320	279
b	Maximum conductor temperature for normal operation	°C	90	90
c	Maximum conductor temperature for short circuit rating	°C	250	250
d	Short circuit rating for 1 sec. duration	K.Amps	17.172	13.5945
3.0	Conductor	---		
a	Material	---	Annealed Plain Copper	Annealed Plain Copper
b	Standard applicable	---	as per IS : 8130-2013 , Class-2	as per IS : 8130-2013 , Class-2
c	Shape of Conductor	---	Stranded, Compacted & Sector Shaped	Stranded, Compacted & Sector Shaped
d	No. of strands	Nos.	19/19	19/7
e	Dia of Strands (Before stranding)	mm	Diameter of strands shall be so choosen to meet the conductor Resistance	Diameter of strands shall be so choosen to meet the conductor Resistance
f	Nominal Cross Sectional Area Phase	sqmm	120	95
g	Nominal Cross Sectional Area Neutral	sqmm	70	50
h	Max. Conductor Resistance Phase at 20°C(DC)	Ω /KM	0.1530	0.193
i	Max. Conductor Resistance Neutral at 20°C(DC)	Ω /KM	0.268	0.387
j	Max. Conductor Resistance Phase at 90°C	Ω /KM	0.19584	0.24704
k	Approximate Reactance of cable at 50 Hz for ref only	Ω /KM	0.073	0.073
l	Approximate Capacitance at 50Hz for ref. only	μ F/Km	0.29	0.29
4.0	Fire Barrier Tape		Glass Mica Tape (Fire Barrier Tape) 2 Tape shall be applied by helical wrapping with 100 % Coverage ,Over conductor throught the length .	

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CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3.5C X 120 SQMM	3.5C X 95 SQMM
5.0	Insulation	---		
a	Material	---	XLPE	XLPE
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Nominal Thickness (Phase)	mm	1.2	1.1
d	Nominal Thickness (Neutral)		1.1	1.0
e	Minimum volume resistivity at 27°C	Ω-cm	1X10 ¹⁴	1X10 ¹⁴
f	Core identification		Red, Yellow, Blue & Black	Red, Yellow, Blue & Black
6.0	Inner Sheath	---		
a	Material	---	Thermoplastic material (PVC with FR Properties)	Thermoplastic material (PVC with FR Properties)
b	Nominal Thickness	mm	1.2	1.2
c	Minimum Thickness	mm	0.76	0.76
d	Extruded/ Wrapped		Extruded	Extruded
e	Colour		Black	Black
7.0	Armouring	---		
a	Material	---	Galvanized Steel	Galvanized Steel
b	Standard applicable	---	IS:3975/1999	IS:3975/1999
c	Type of Armour		Round Wire	Round Wire
d	Size of armour	mm	2.0 ± 0.050	2.0 ± 0.050
e	Binder Tape Over Armour		Synthetic tape may be applied by helical wrapping with 100 % coverage over armour	
8.0	Outer Sheath	---		
a	Material	---	LSZH	LSZH
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Minimum Thickness	mm	1.88	1.72
d	Nominal Thickness	mm	2.6	2.4
e	Overall diameter	mm	45	41
f	Tolerance in overall diameter	± mm	± 4	± 4
g	Outer Sheath colour		Black	Black
8.1	Circuit Integrity Test		Shall pass the test Categories C as per IS:17505 (Part-1) :2021 at 950 ±40 0 C for 3 Hours Category F	
9.0	Min. Bending Radius		12 X Cable Dia	12 X Cable Dia
10.0	Packing Detail			
a	Type of Packing Drum		Wooden Drum	Wooden Drum
b	Packing Length with Tolerance	metres	500 ± 5%	500 ± 5%
c	Order Quantity Tolerance		± 5%	± 5%
11.0	Marking on Outer Sheath			
a	By Printing		RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No	RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No
b	Sequential Length Marking		Shall be provided at every metere by printing	Shall be provided at every metere by printing

Note: 5 % of Ordered quantity may be supplied in Assorted Non-Standard length but shall not be less than 50 Metres

Note: In case of any dispute/typographical error interpretation of the IS specification shall prevail.

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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3.5C X 70 SQMM	3.5C X 50 SQMM
1.0	General			
a	Brand Name	---	RALLISON	RALLISON
b	Standards applicable	---	IS: 17505 (Part 1) :2021 with up to date amendment	IS: 17505 (Part 1) :2021 with up to date amendment
c	Voltage Grade	Volts	1100 Volts	1100 Volts
d	Permissible voltage & frequency variation	%	Voltage variation $\pm$ 10%, Frequency variation $\pm$ 5%	Voltage variation $\pm$ 10%, Frequency variation $\pm$ 5%
e	Suitable for earthed / unearthed systems		Suitable for earth and un earthed system	Suitable for earth and un earthed system
2.0	Current rating			
a	Continuous current rating under normal conditions			
i	In Ground at 30°C	Amps	212	173
ii	In Air at 40°C	Amps	226	179
b	Maximum conductor temperature for normal operation	°C	90	90
c	Maximum conductor temperature for short circuit rating	°C	250	250
d	Short circuit rating for 1 sec. duration	K.Amps	10.017	7.155
3.0	Conductor	---		
a	Material	---	Annealed Plain Copper	Annealed Plain Copper
b	Standard applicable	---	as per IS : 8130-2013 , Class-2	as per IS : 8130-2013 , Class-2
c	Shape of Conductor	---	Stranded, Compacted & Sector Shaped	Stranded, Compacted & Sector Shaped
d	No. of strands	Nos.	19/7	7/7
e	Dia of Strands (Before stranding)	mm	Diameter of strands shall be so choosen to meet the conductor Resistance	Diameter of strands shall be so choosen to meet the conductor Resistance
f	Nominal Cross Sectional Area Phase	sqmm	70	50
g	Nominal Cross Sectional Area Neutral	sqmm	35	25
h	Max. Conductor Resistance Phase at 20°C(DC)	Ω/KM	0.2680	0.387
i	Max. Conductor Resistance Neutral at 20°C(DC)	Ω/KM	0.524	0.727
j	Max. Conductor Resistance Phase at 90°C	Ω/KM	0.34304	0.49536
k	Approximate Reactance of cable at 50 Hz for ref only	Ω/KM	0.075	0.076
l	Approximate Capacitance at 50Hz for ref. only	μF/Km	0.26	0.24
4.0	Fire Barrier Tape		Glass Mica Tape (Fire Barrier Tape) 2 Tape shall be applied by helical wrapping with 100 % Coverage ,Over conductor throught the length .	

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CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3.5C X 70 SQMM	3.5C X 50 SQMM
5.0	Insulation	---		
a	Material	---	XLPE	XLPE
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Nominal Thickness (Phase)	mm	1.1	1
d	Nominal Thickness (Neutral)		0.9	0.9
e	Minimum volume resistivity at 27°C	Ω-cm	1X10 ¹⁴	1X10 ¹⁴
f	Core identification		Red, Yellow, Blue & Black	Red, Yellow, Blue & Black
6.0	Inner Sheath	---		
a	Material	---	Thermoplastic material (PVC with FR Properties)	Thermoplastic material (PVC with FR Properties)
b	Nominal Thickness	mm	1.2	1.0
c	Minimum Thickness	mm	0.76	0.6
d	Extruded/ Wrapped		Extruded	Extruded
e	Colour		Black	Black
7.0	Armouring	---		
a	Material	---	Galvanized Steel	Galvanized Steel
b	Standard applicable	---	IS:3975/1999	IS:3975/1999
c	Type of Armour		Round Wire	Round Wire
d	Size of armour	mm	2.0 ± 0.050	1.60 ± 0.045
e	Binder Tape Over Armour		Synthetic tape may be applied by helical wrapping with 100 % coverage over armour	
8.0	Outer Sheath	---		
a	Material	---	LSZH	LSZH
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Minimum Thickness	mm	1.56	1.56
d	Nominal Thickness	mm	2.2	2.2
e	Overall diameter	mm	37	32
f	Tolerance in overall diameter	± mm	± 4	± 4
g	Outer Sheath colour		Black	Black
8.1	Circuit Integrity Test		Shall pass the test Categories C as per IS:17505 (Part-1) :2021 at 950 ±40 0 C for 3 Hours Category F	
9.0	Min. Bending Radius		12 X Cable Dia	12 X Cable Dia
10.0	Packing Detail			
a	Type of Packing Drum		Wooden Drum	Wooden Drum
b	Packing Length with Tolerance	metres	1000/500 ± 5%	1000 / 500 ± 5 %
c	Order Quantity Tolerance		± 5%	± 5%
11.0	Marking on Outer Sheath			
a	By Printing		RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No	RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No
b	Sequential Length Marking		Shall be provided at every metere by printing	Shall be provided at every metere by printing

Note: 5 % of Ordered quantity may be supplied in Assorted Non-Standard length but shall not be less than 50 Metres

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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3.5C X 35 SQMM	3.5C X 25 SQMM
1.0	General			
a	Brand Name	---	RALLISON	RALLISON
b	Standards applicable	---	IS: 17505 (Part 1) :2021 with up to date amendment	IS: 17505 (Part 1) :2021 with up to date amendment
c	Voltage Grade	Volts	1100 Volts	1100 Volts
d	Permissible voltage & frequency variation	%	Voltage variation $\pm$ 10%, Frequency variation $\pm$ 5%	Voltage variation $\pm$ 10%, Frequency variation $\pm$ 5%
e	Suitable for earthed / unearthed systems		Suitable for earth and un earthed system	Suitable for earth and un earthed system
2.0	Current rating			
a	Continuous current rating under normal conditions			
i	In Ground at 30°C	Amps	146	122
ii	In Air at 40°C	Amps	147	119
b	Maximum conductor temperature for normal operation	°C	90	90
c	Maximum conductor temperature for short circuit rating	°C	250	250
d	Short circuit rating for 1 sec. duration	K.Amps	5.0085	3.5775
3.0	Conductor	---		
a	Material	---	Annealed Plain Copper	Annealed Plain Copper
b	Standard applicable	---	as per IS : 8130-2013 , Class-2	as per IS : 8130-2013 , Class-2
c	Shape of Conductor	---	Stranded, Compacted & Sector Shaped	Stranded, Compacted & Sector Shaped
d	No. of strands	Nos.	7/7	7/7
e	Dia of Strands (Before stranding)	mm	Diameter of strands shall be so choosen to meet the conductor Resistance	Diameter of strands shall be so choosen to meet the conductor Resistance
f	Nominal Cross Sectional Area Phase	sqmm	35	25
g	Nominal Cross Sectional Area Neutral	sqmm	16	16
h	Max. Conductor Resistance Phase at 20°C(DC)	Ω /KM	0.524	0.727
i	Max. Conductor Resistance Neutral at 20°C(DC)	Ω /KM	1.15	1.15
j	Max. Conductor Resistance Phase at 90°C	Ω /KM	0.67072	0.93056
k	Approximate Reactance of cable at 50 Hz for ref only	Ω /KM	0.077	0.079
l	Approximate Capacitance at 50Hz for ref. only	μ F/Km	0.23	0.20
4.0	Fire Barrier Tape		Glass Mica Tape (Fire Barrier Tape) 2 Tape shall be applied by helical wrapping with 100 % Coverage ,Over conductor throught the length .	

LKB ENGINEERING PRIVATE LIMITED
E-418 TO E-421 , CHOPANKI INDUSTRIAL AREA , BHIWADI ,
DISTT. ALWAR , RAJASTHAN-301019

(AN ISO CERTIFIED , NABL ACCREDITED AND BIS CERTIFIED LABORATORY)

TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3.5C X 35 SQMM	3.5C X 25 SQMM
5.0	Insulation	---		
a	Material	---	XLPE	XLPE
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Nominal Thickness (Phase)	mm	0.9	0.9
d	Nominal Thickness (Neutral)		0.7	0.7
e	Minimum volume resistivity at 27°C	Ω-cm	1X10 ¹⁴	1X10 ¹⁴
f	Core identification		Red, Yellow, Blue & Black	Red, Yellow, Blue & Black
6.0	Inner Sheath	---		
a	Material	---	Thermoplastic material (PVC with FR Properties)	Thermoplastic material (PVC with FR Properties)
b	Nominal Thickness	mm	1	1.0
c	Minimum Thickness	mm	0.6	0.6
d	Extruded/ Wrapped		Extruded	Extruded
e	Colour		Black	Black
7.0	Armouring	---		
a	Material	---	Galvanized Steel	Galvanized Steel
b	Standard applicable	---	IS:3975/1999	IS:3975/1999
c	Type of Armour		Round Wire	Round Wire
d	Size of armour	mm	1.60 ± 0.045	1.60 ± 0.045
e	Binder Tape Over Armour		Synthetic tape may be applied by helical wrapping with 100 % coverage over armour	
8.0	Outer Sheath	---		
a	Material	---	LSZH	LSZH
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Minimum Thickness	mm	1.4	1.4
d	Nominal Thickness	mm	2.0	2.0
e	Overall diameter	mm	28	26
f	Tolerance in overall diameter	± mm	± 4	± 4
g	Outer Sheath colour		Black	Black
8.1	Circuit Integrity Test		Shall pass the test Categories C as per IS:17505 (Part-1) :2021 at 950 ±40 0 C for 3 Hours Category F	
9.0	Min. Bending Radius		12 X Cable Dia	12 X Cable Dia
10.0	Packing Detail			
a	Type of Packing Drum		Wooden Drum	Wooden Drum
b	Packing Length with Tolerance	metres	1000/500 ± 5%	1000 / 500 ± 5 %
c	Order Quantity Tolerance		± 5%	± 5%
11.0	Marking on Outer Sheath			
a	By Printing		RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No	RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No
b	Sequential Length Marking		Shall be provided at every metere by printing	Shall be provided at every metere by printing

Note: 5 % of Ordered quantity may be supplied in Assorted Non-Standard length but shall not be less than 50 Metres

Note: In case of any dispute/typographical error interpretation of the IS specification shall prevail.

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(AN ISO CERTIFIED , NABL ACCREDITED AND BIS CERTIFIED LABORATORY)

TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 300 SQMM	3C X 240 SQMM
1.0	General			
a	Brand Name	---	RALLISON	RALLISON
b	Standards applicable	---	IS: 17505 (Part 1) :2021 with up to date amendment	IS: 17505 (Part 1) :2021 with up to date amendment
c	Voltage Grade	Volts	1100 Volts	1100 Volts
d	Permissible voltage & frequency variation	%	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$
e	Suitable for earthed / unearthed systems		Suitable for earth and un earthed system	Suitable for earth and un earthed system
2.0	Current rating			
a	Continuous current rating under normal conditions			
i	In Ground at 30°C	Amps	469	418
ii	In Air at 40°C	Amps	574	500
b	Maximum conductor temperature for normal operation	°C	90	90
c	Maximum conductor temperature for short circuit rating	°C	250	250
d	Short circuit rating for 1 sec. duration	K.Amps	42.93	34.344
3.0	Conductor	---		
a	Material	---	Annealed Plain Copper	Annealed Plain Copper
b	Standard applicable	---	as per IS : 8130-2013 , Class-2	as per IS : 8130-2013 , Class-2
c	Shape of Conductor	---	Stranded, Compacted & Sector Shaped	Stranded, Compacted & Sector Shaped
d	No. of strands	Nos.	37	37
e	Dia of Strands (Before stranding)	mm	Diameter of strands shall be so choosen to meet the conductor Resistance	Diameter of strands shall be so choosen to meet the conductor Resistance
f	Nominal Cross Sectional Area Phase	sqmm	300	240
g	Nominal Cross Sectional Area Neutral	sqmm	NA	NA
h	Max. Conductor Resistance Phase at 20°C(DC)	Ω /KM	0.0601	0.075
i	Max. Conductor Resistance Neutral at 20°C(DC)	Ω /KM	NA	NA
j	Max. Conductor Resistance Phase at 90°C	Ω /KM	0.076928	0.096512
k	Approximate Reactance of cable at 50 Hz for ref only	Ω /KM	0.072	0.073
l	Approximate Capacitance at 50Hz for ref. only	μ F/Km	0.33	0.31
4.0	Fire Barrier Tape		Glass Mica Tape (Fire Barrier Tape) 2 Tape shall be applied by helical wrapping with 100 % Coverage ,Over conductor throught the length .	

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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 300 SQMM	3C X 240 SQMM
5.0	Insulation	---		
a	Material	---	XLPE	XLPE
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Nominal Thickness (Phase)	mm	1.8	1.7
d	Nominal Thickness (Neutral)		N/A	N/A
e	Minimum volume resistivity at 27°C	Ω-cm	1X10 ¹⁴	1X10 ¹⁴
f	Core identification		Red, Yellow & Blue	Red, Yellow & Blue
6.0	Inner Sheath	---		
a	Material	---	Thermoplastic material (PVC with FR Properties)	Thermoplastic material (PVC with FR Properties)
b	Nominal Thickness	mm	1.6	1.6
c	Minimum Thickness	mm	1.08	1.08
d	Extruded/ Wrapped		Extruded	Extruded
e	Colour		Black	Black
7.0	Armouring	---		
a	Material	---	Galvanized Steel	Galvanized Steel
b	Standard applicable	---	IS:3975/1999	IS:3975/1999
c	Type of Armour		Round Wire	Round Wire
d	Size of armour	mm	2.5 ± 0.065	2.5 ± 0.065
e	Binder Tape Over Armour		Synthetic tape may be applied by helical wrapping with 100 % coverage over armour	
8.0	Outer Sheath	---		
a	Material	---	LSZH	LSZH
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Minimum Thickness	mm	2.36	2.2
d	Nominal Thickness	mm	3.2	3.0
e	Overall diameter	mm	65	59
f	Tolerance in overall diameter	± mm	± 4	± 4
g	Outer Sheath colour		Black	Black
8.1	Circuit Integrity Test		Shall pass the test Categories C as per IS:17505 (Part-1) :2021 at 950 ±40 0 C for 3 Hours Category F	
9.0	Min. Bending Radius		12 X Cable Dia	12 X Cable Dia
10.0	Packing Detail			
a	Type of Packing Drum		Wooden Drum	Wooden Drum
b	Packing Length with Tolerance	metres	250 ± 5%	250 ± 5%
c	Order Quantity Tolerance		± 5%	± 5%
11.0	Marking on Outer Sheath			
a	By Printing		RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No	RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No
b	Sequential Length Marking		Shall be provided at every metere by printing	Shall be provided at every metere by printing

Note: 5 % of Ordered quantity may be supplied in Assorted Non-Standard length but shall not be less than 50 Metres

Note: In case of any dispute/typographical error interpretation of the IS specification shall prevail.

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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 185 SQMM	3C X 150 SQMM
1.0	General			
a	Brand Name	---	RALLISON	RALLISON
b	Standards applicable	---	IS: 17505 (Part 1) :2021 with up to date amendment	IS: 17505 (Part 1) :2021 with up to date amendment
c	Voltage Grade	Volts	1100 Volts	1100 Volts
d	Permissible voltage & frequency variation	%	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$
e	Suitable for earthed / unearthed systems		Suitable for earth and un earthed system	Suitable for earth and un earthed system
2.0	Current rating			
a	Continuous current rating under normal conditions			
i	In Ground at 30°C	Amps	362	321
ii	In Air at 40°C	Amps	422	365
b	Maximum conductor temperature for normal operation	°C	90	90
c	Maximum conductor temperature for short circuit rating	°C	250	250
d	Short circuit rating for 1 sec. duration	K.Amps	26.4735	21.465
3.0	Conductor	---		
a	Material	---	Annealed Plain Copper	Annealed Plain Copper
b	Standard applicable	---	as per IS : 8130-2013 , Class-2	as per IS : 8130-2013 , Class-2
c	Shape of Conductor	---	Stranded, Compacted & Sector Shaped	Stranded, Compacted & Sector Shaped
d	No. of strands	Nos.	37	19
e	Dia of Strands (Before stranding)	mm	Diameter of strands shall be so choosen to meet the conductor Resistance	Diameter of strands shall be so choosen to meet the conductor Resistance
f	Nominal Cross Sectional Area Phase	sqmm	185	150
g	Nominal Cross Sectional Area Neutral	sqmm	NA	NA
h	Max. Conductor Resistance Phase at 20°C(DC)	Ω /KM	0.0991	0.124
i	Max. Conductor Resistance Neutral at 20°C(DC)	Ω /KM	NA	NA
j	Max. Conductor Resistance Phase at 90°C	Ω /KM	0.126848	0.15872
k	Approximate Reactance of cable at 50 Hz for ref only	Ω /KM	0.073	0.073
l	Approximate Capacitance at 50Hz for ref. only	μ F/Km	0.29	0.29
4.0	Fire Barrier Tape		Glass Mica Tape (Fire Barrier Tape) 2 Tape shall be applied by helical wrapping with 100 % Coverage ,Over conductor throught the length .	

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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 185 SQMM	3C X 150 SQMM
5.0	Insulation	---		
a	Material	---	XLPE	XLPE
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Nominal Thickness (Phase)	mm	1.6	1.4
d	Nominal Thickness (Neutral)		N/A	N/A
e	Minimum volume resistivity at 27°C	Ω-cm	1X10 ¹⁴	1X10 ¹⁴
f	Core identification		Red, Yellow & Blue	Red, Yellow & Blue
6.0	Inner Sheath	---		
a	Material	---	Thermoplastic material (PVC with FR Properties)	Thermoplastic material (PVC with FR Properties)
b	Nominal Thickness	mm	1.4	1.4
c	Minimum Thickness	mm	0.92	0.92
d	Extruded/ Wrapped		Extruded	Extruded
e	Colour		Black	Black
7.0	Armouring	---		
a	Material	---	Galvanized Steel	Galvanized Steel
b	Standard applicable	---	IS:3975/1999	IS:3975/1999
c	Type of Armour		Round Wire	Round Wire
d	Size of armour	mm	2.50 ± 0.065	2.0 ± 0.050
e	Binder Tape Over Armour		Synthetic tape may be applied by helical wrapping with 100 % coverage over armour	
8.0	Outer Sheath	---		
a	Material	---	LSZH	LSZH
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Minimum Thickness	mm	2.2	1.88
d	Nominal Thickness	mm	3.0	2.6
e	Overall diameter	mm	54	47
f	Tolerance in overall diameter	± mm	± 4	± 4
g	Outer Sheath colour		Black	Black
8.1	Circuit Integrity Test		Shall pass the test Categories C as per IS:17505 (Part-1) :2021 at 950 ±40 0 C for 3 Hours Category F	
9.0	Min. Bending Radius		12 X Cable Dia	12 X Cable Dia
10.0	Packing Detail			
a	Type of Packing Drum		Wooden Drum	Wooden Drum
b	Packing Length with Tolerance	metres	250 ± 5%	500 ± 5%
c	Order Quantity Tolerance		± 5%	± 5%
11.0	Marking on Outer Sheath			
a	By Printing		RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No	RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No
b	Sequential Length Marking		Shall be provided at every metere by printing	Shall be provided at every metere by printing

Note: 5 % of Ordered quantity may be supplied in Assorted Non-Standard length but shall not be less than 50 Metres

Note: In case of any dispute/typographical error interpretation of the IS specification shall prevail.

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E-418 TO E-421 , CHOPANKI INDUSTRIAL AREA , BHIWADI ,
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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 120 SQMM	3C X 95 SQMM
1.0	General			
a	Brand Name	---	RALLISON	RALLISON
b	Standards applicable	---	IS: 17505 (Part 1) :2021 with up to date amendment	IS: 17505 (Part 1) :2021 with up to date amendment
c	Voltage Grade	Volts	1100 Volts	1100 Volts
d	Permissible voltage & frequency variation	%	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$
e	Suitable for earthed / unearthed systems		Suitable for earth and un earthed system	Suitable for earth and un earthed system
2.0	Current rating			
a	Continuous current rating under normal conditions			
i	In Ground at 30°C	Amps	287	254
ii	In Air at 40°C	Amps	320	279
b	Maximum conductor temperature for normal operation	°C	90	90
c	Maximum conductor temperature for short circuit rating	°C	250	250
d	Short circuit rating for 1 sec. duration	K.Amps	17.172	13.5945
3.0	Conductor	---		
a	Material	---	Annealed Plain Copper	Annealed Plain Copper
b	Standard applicable	---	as per IS : 8130-2013 , Class-2	as per IS : 8130-2013 , Class-2
c	Shape of Conductor	---	Stranded, Compacted & Sector Shaped	Stranded, Compacted & Sector Shaped
d	No. of strands	Nos.	19	19
e	Dia of Strands (Before stranding)	mm	Diameter of strands shall be so choosen to meet the conductor Resistance	Diameter of strands shall be so choosen to meet the conductor Resistance
f	Nominal Cross Sectional Area Phase	sqmm	120	95
g	Nominal Cross Sectional Area Neutral	sqmm	NA	NA
h	Max. Conductor Resistance Phase at 20°C(DC)	Ω /KM	0.1530	0.193
i	Max. Conductor Resistance Neutral at 20°C(DC)	Ω /KM	NA	NA
j	Max. Conductor Resistance Phase at 90°C	Ω /KM	0.19584	0.24704
k	Approximate Reactance of cable at 50 Hz for ref only	Ω /KM	0.073	0.073
l	Approximate Capacitance at 50Hz for ref. only	μ F/Km	0.29	0.29
4.0	Fire Barrier Tape		Glass Mica Tape (Fire Barrier Tape) 2 Tape shall be applied by helical wrapping with 100 % Coverage ,Over conductor throught the length .	

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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 120 SQMM	3C X 95 SQMM
5.0	Insulation	---		
a	Material	---	XLPE	XLPE
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Nominal Thickness (Phase)	mm	1.2	1.1
d	Nominal Thickness (Neutral)		N/A	N/A
e	Minimum volume resistivity at 27°C	Ω-cm	1X10 ¹⁴	1X10 ¹⁴
f	Core identification		Red, Yellow & Blue	Red, Yellow & Blue
6.0	Inner Sheath	---		
a	Material	---	Thermoplastic material (PVC with FR Properties)	Thermoplastic material (PVC with FR Properties)
b	Nominal Thickness	mm	1.2	1.2
c	Minimum Thickness	mm	0.76	0.76
d	Extruded/ Wrapped		Extruded	Extruded
e	Colour		Black	Black
7.0	Armouring	---		
a	Material	---	Galvanized Steel	Galvanized Steel
b	Standard applicable	---	IS:3975/1999	IS:3975/1999
c	Type of Armour		Round Wire	Round Wire
d	Size of armour	mm	2.0 ± 0.050	2.0 ± 0.050
e	Binder Tape Over Armour		Synthetic tape may be applied by helical wrapping with 100 % coverage over armour	
8.0	Outer Sheath	---		
a	Material	---	LSZH	LSZH
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Minimum Thickness	mm	1.72	1.56
d	Nominal Thickness	mm	2.4	2.2
e	Overall diameter	mm	43	39
f	Tolerance in overall diameter	± mm	± 4	± 4
g	Outer Sheath colour		Black	Black
8.1	Circuit Integrity Test		Shall pass the test Categories C as per IS:17505 (Part-1) :2021 at 950 ±40 0 C for 3 Hours Category F	
9.0	Min. Bending Radius		12 X Cable Dia	12 X Cable Dia
10.0	Packing Detail			
a	Type of Packing Drum		Wooden Drum	Wooden Drum
b	Packing Length with Tolerance	metres	500 ± 5%	500 ± 5%
c	Order Quantity Tolerance		± 5%	± 5%
11.0	Marking on Outer Sheath			
a	By Printing		RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No	RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No
b	Sequential Length Marking		Shall be provided at every metere by printing	Shall be provided at every metere by printing

Note: 5 % of Ordered quantity may be supplied in Assorted Non-Standard length but shall not be less than 50 Metres

Note: In case of any dispute/typographical error interpretation of the IS specification shall prevail.

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E-418 TO E-421 , CHOPANKI INDUSTRIAL AREA , BHIWADI ,
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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 70 SQMM	3C X 50 SQMM
1.0	General			
a	Brand Name	---	RALLISON	RALLISON
b	Standards applicable	---	IS: 17505 (Part 1) :2021 with up to date amendment	IS: 17505 (Part 1) :2021 with up to date amendment
c	Voltage Grade	Volts	1100 Volts	1100 Volts
d	Permissible voltage & frequency variation	%	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$
e	Suitable for earthed / unearthed systems		Suitable for earth and un earthed system	Suitable for earth and un earthed system
2.0	Current rating			
a	Continuous current rating under normal conditions			
i	In Ground at 30°C	Amps	212	173
ii	In Air at 40°C	Amps	226	179
b	Maximum conductor temperature for normal operation	°C	90	90
c	Maximum conductor temperature for short circuit rating	°C	250	250
d	Short circuit rating for 1 sec. duration	K.Amps	10.017	7.155
3.0	Conductor	---		
a	Material	---	Annealed Plain Copper	Annealed Plain Copper
b	Standard applicable	---	as per IS : 8130-2013 , Class-2	as per IS : 8130-2013 , Class-2
c	Shape of Conductor	---	Stranded, Compacted & Sector Shaped	Stranded, Compacted & Sector Shaped
d	No. of strands	Nos.	19	7
e	Dia of Strands (Before stranding)	mm	Diameter of strands shall be so choosen to meet the conductor Resistance	Diameter of strands shall be so choosen to meet the conductor Resistance
f	Nominal Cross Sectional Area Phase	sqmm	70	50
g	Nominal Cross Sectional Area Neutral	sqmm	NA	NA
h	Max. Conductor Resistance Phase at 20°C(DC)	Ω /KM	0.268	0.387
i	Max. Conductor Resistance Neutral at 20°C(DC)	Ω /KM	NA	NA
j	Max. Conductor Resistance Phase at 90°C	Ω /KM	0.34304	0.49536
k	Approximate Reactance of cable at 50 Hz for ref only	Ω /KM	0.075	0.076
l	Approximate Capacitance at 50Hz for ref. only	μ F/Km	0.26	0.24
4.0	Fire Barrier Tape		Glass Mica Tape (Fire Barrier Tape) 2 Tape shall be applied by helical wrapping with 100 % Coverage ,Over conductor throught the length .	

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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 70 SQMM	3C X 50 SQMM
5.0	Insulation	---		
a	Material	---	XLPE	XLPE
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Nominal Thickness (Phase)	mm	1.1	1
d	Nominal Thickness (Neutral)		N/A	N/A
e	Minimum volume resistivity at 27°C	Ω-cm	1X10 ¹⁴	1X10 ¹⁴
f	Core identification		Red, Yellow & Blue	Red, Yellow & Blue
6.0	Inner Sheath	---		
a	Material	---	Thermoplastic material (PVC with FR Properties)	Thermoplastic material (PVC with FR Properties)
b	Nominal Thickness	mm	1.2	1.0
c	Minimum Thickness	mm	0.76	0.6
d	Extruded/ Wrapped		Extruded	Extruded
e	Colour		Black	Black
7.0	Armouring	---		
a	Material	---	Galvanized Steel	Galvanized Steel
b	Standard applicable	---	IS:3975/1999	IS:3975/1999
c	Type of Armour		Round Wire	Round Wire
d	Size of armour	mm	2.0 ± 0.050	1.60 ± 0.045
e	Binder Tape Over Armour		Synthetic tape may be applied by helical wrapping with 100 % coverage over armour	
8.0	Outer Sheath	---		
a	Material	---	LSZH	LSZH
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Minimum Thickness	mm	1.56	1.56
d	Nominal Thickness	mm	2.2	2.2
e	Overall diameter	mm	36	31
f	Tolerance in overall diameter	± mm	± 4	± 4
g	Outer Sheath colour		Black	Black
8.1	Circuit Integrity Test		Shall pass the test Categories C as per IS:17505 (Part-1) :2021 at 950 ±40 0 C for 3 Hours Category F	
9.0	Min. Bending Radius		12 X Cable Dia	12 X Cable Dia
10.0	Packing Detail			
a	Type of Packing Drum		Wooden Drum	Wooden Drum
b	Packing Length with Tolerance	metres	1000/500 ± 5%	1000 / 500 ± 5 %
c	Order Quantity Tolerance		± 5%	± 5%
11.0	Marking on Outer Sheath			
a	By Printing		RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No	RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No
b	Sequential Length Marking		Shall be provided at every metere by printing	Shall be provided at every metere by printing

Note: 5 % of Ordered quantity may be supplied in Assorted Non-Standard length but shall not be less than 50 Metres

Note: In case of any dispute/typographical error interpretation of the IS specification shall prevail.

LKB ENGINEERING PRIVATE LIMITED
E-418 TO E-421 , CHOPANKI INDUSTRIAL AREA , BHIWADI ,
DISTT. ALWAR , RAJASTHAN-301019

(AN ISO CERTIFIED , NABL ACCREDITED AND BIS CERTIFIED LABORATORY)

TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 35 SQMM	3C X 25 SQMM
1.0	General			
a	Brand Name	---	RALLISON	RALLISON
b	Standards applicable	---	IS: 17505 (Part 1) :2021 with up to date amendment	IS: 17505 (Part 1) :2021 with up to date amendment
c	Voltage Grade	Volts	1100 Volts	1100 Volts
d	Permissible voltage & frequency variation	%	Voltage variation $\pm$ 10%, Frequency variation $\pm$ 5%	Voltage variation $\pm$ 10%, Frequency variation $\pm$ 5%
e	Suitable for earthed / unearthed systems		Suitable for earth and un earthed system	Suitable for earth and un earthed system
2.0	Current rating			
a	Continuous current rating under normal conditions			
i	In Ground at 30°C	Amps	146	122
ii	In Air at 40°C	Amps	147	119
b	Maximum conductor temperature for normal operation	°C	90	90
c	Maximum conductor temperature for short circuit rating	°C	250	250
d	Short circuit rating for 1 sec. duration	K.Amps	5.0085	3.5775
3.0	Conductor	---		
a	Material	---	Annealed Plain Copper	Annealed Plain Copper
b	Standard applicable	---	as per IS : 8130-2013 , Class-2	as per IS : 8130-2013 , Class-2
c	Shape of Conductor	---	Stranded, Compacted & Sector Shaped	Stranded, Compacted & Sector Shaped
d	No. of strands	Nos.	7	7
e	Dia of Strands (Before stranding)	mm	Diameter of strands shall be so choosen to meet the conductor Resistance	Diameter of strands shall be so choosen to meet the conductor Resistance
f	Nominal Cross Sectional Area Phase	sqmm	35	25
g	Nominal Cross Sectional Area Neutral	sqmm	NA	NA
h	Max. Conductor Resistance Phase at 20°C(DC)	Ω/KM	0.524	0.727
i	Max. Conductor Resistance Neutral at 20°C(DC)	Ω/KM	NA	NA
j	Max. Conductor Resistance Phase at 90°C	Ω/KM	0.67072	0.93056
k	Approximate Reactance of cable at 50 Hz for ref only	Ω/KM	0.077	0.079
l	Approximate Capacitance at 50Hz for ref. only	μF/Km	0.23	0.20
4.0	Fire Barrier Tape		Glass Mica Tape (Fire Barrier Tape) 2 Tape shall be applied by helical wrapping with 100 % Coverage ,Over conductor throught the length .	

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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 35 SQMM	3C X 25 SQMM
5.0	Insulation	---		
a	Material	---	XLPE	XLPE
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Nominal Thickness (Phase)	mm	0.9	0.9
d	Nominal Thickness (Neutral)		N/A	N/A
e	Minimum volume resistivity at 27°C	Ω-cm	1X10 ¹⁴	1X10 ¹⁴
f	Core identification		Red, Yellow & Blue	Red, Yellow & Blue
6.0	Inner Sheath	---		
a	Material	---	Thermoplastic material (PVC with FR Properties)	Thermoplastic material (PVC with FR Properties)
b	Nominal Thickness	mm	1	1.0
c	Minimum Thickness	mm	0.6	0.6
d	Extruded/ Wrapped		Extruded	Extruded
e	Colour		Black	Black
7.0	Armouring	---		
a	Material	---	Galvanized Steel	Galvanized Steel
b	Standard applicable	---	IS:3975/1999	IS:3975/1999
c	Type of Armour		Round Wire	Round Wire
d	Size of armour	mm	1.60 ± 0.045	1.60 ± 0.045
e	Binder Tape Over Armour		Synthetic tape may be applied by helical wrapping with 100 % coverage over armour	
8.0	Outer Sheath	---		
a	Material	---	LSZH	LSZH
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Minimum Thickness	mm	1.4	1.4
d	Nominal Thickness	mm	2.0	2.0
e	Overall diameter	mm	27	25
f	Tolerance in overall diameter	± mm	± 4	± 4
g	Outer Sheath colour		Black	Black
8.1	Circuit Integrity Test		Shall pass the test Categories C as per IS:17505 (Part-1) :2021 at 950 ±40 0 C for 3 Hours Category F	
9.0	Min. Bending Radius		12 X Cable Dia	12 X Cable Dia
10.0	Packing Detail			
a	Type of Packing Drum		Wooden Drum	Wooden Drum
b	Packing Length with Tolerance	metres	1000/500 ± 5%	1000 / 500 ± 5 %
c	Order Quantity Tolerance		± 5%	± 5%
11.0	Marking on Outer Sheath			
a	By Printing		RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No	RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No
b	Sequential Length Marking		Shall be provided at every metere by printing	Shall be provided at every metere by printing

Note: 5 % of Ordered quantity may be supplied in Assorted Non-Standard length but shall not be less than 50 Metres

Note: In case of any dispute/typographical error interpretation of the IS specification shall prevail.

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E-418 TO E-421 , CHOPANKI INDUSTRIAL AREA , BHIWADI ,
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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 16 SQMM	3C X 10 SQMM
1.0	General			
a	Brand Name	---	RALLISON	RALLISON
b	Standards applicable	---	IS: 17505 (Part 1) :2021 with up to date amendment	IS: 17505 (Part 1) :2021 with up to date amendment
c	Voltage Grade	Volts	1100 Volts	1100 Volts
d	Permissible voltage & frequency variation	%	Voltage variation $\pm$ 10%, Frequency variation $\pm$ 5%	Voltage variation $\pm$ 10%, Frequency variation $\pm$ 5%
e	Suitable for earthed / unearthed systems		Suitable for earth and un earthed system	Suitable for earth and un earthed system
2.0	Current rating			
a	Continuous current rating under normal conditions			
i	In Ground at 30°C	Amps	95	74
ii	In Air at 40°C	Amps	89	70
b	Maximum conductor temperature for normal operation	°C	90	90
c	Maximum conductor temperature for short circuit rating	°C	250	250
d	Short circuit rating for 1 sec. duration	K.Amps	2.2896	1.431
3.0	Conductor	---		
a	Material	---	Annealed Plain Copper	Annealed Plain Copper
b	Standard applicable	---	as per IS : 8130-2013 , Class-2	as per IS : 8130-2013 , Class-2
c	Shape of Conductor	---	Stranded, Compacted & Sector Shaped	Stranded, Circular
d	No. of strands	Nos.	7	7
e	Dia of Strands (Before stranding)	mm	Diameter of strands shall be so choosen to meet the conductor Resistance	Diameter of strands shall be so choosen to meet the conductor Resistance
f	Nominal Cross Sectional Area Phase	sqmm	16	10
g	Nominal Cross Sectional Area Neutral	sqmm	NA	NA
h	Max. Conductor Resistance Phase at 20°C(DC)	Ω /KM	1.15	1.83
i	Max. Conductor Resistance Neutral at 20°C(DC)	Ω /KM	NA	NA
j	Max. Conductor Resistance Phase at 90°C	Ω /KM	1.472	2.3424
k	Approximate Reactance of cable at 50 Hz for ref only	Ω /KM	0.081	0.083
l	Approximate Capacitance at 50Hz for ref. only	μ F/Km	0.18	0.16
4.0	Fire Barrier Tape		Glass Mica Tape (Fire Barrier Tape) 2 Tape shall be applied by helical wrapping with 100 % Coverage ,Over conductor throught the length .	

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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 16 SQMM	3C X 10 SQMM
5.0	Insulation	---		
a	Material	---	XLPE	XLPE
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Nominal Thickness (Phase)	mm	0.7	0.7
d	Nominal Thickness (Neutral)		N/A	N/A
e	Minimum volume resistivity at 27°C	Ω-cm	1X10 ¹⁴	1X10 ¹⁴
f	Core identification		Red, Yellow & Blue	Red, Yellow & Blue
6.0	Inner Sheath	---		
a	Material	---	Thermoplastic material (PVC with FR Properties)	Thermoplastic material (PVC with FR Properties)
b	Nominal Thickness	mm	1	1.0
c	Minimum Thickness	mm	0.6	0.6
d	Extruded/ Wrapped		Extruded	Extruded
e	Colour		Black	Black
7.0	Armouring	---		
a	Material	---	Galvanized Steel	Galvanized Steel
b	Standard applicable	---	IS:3975/1999	IS:3975/1999
c	Type of Armour		Round Wire	Round Wire
d	Size of armour	mm	1.60 ± 0.045	1.40 ± 0.040
e	Binder Tape Over Armour		Synthetic tape may be applied by helical wrapping with 100 % coverage over armour	
8.0	Outer Sheath	---		
a	Material	---	LSZH	LSZH
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Minimum Thickness	mm	1.4	1.4
d	Nominal Thickness	mm	2.0	2.0
e	Overall diameter	mm	22	20
f	Tolerance in overall diameter	± mm	± 4	± 4
g	Outer Sheath colour		Black	Black
8.1	Circuit Integrity Test		Shall pass the test Categories C as per IS:17505 (Part-1) :2021 at 950 ±40 0 C for 3 Hours Category F	
9.0	Min. Bending Radius		12 X Cable Dia	12 X Cable Dia
10.0	Packing Detail			
a	Type of Packing Drum		Wooden Drum	Wooden Drum
b	Packing Length with Tolerance	metres	1000/500 ± 5%	1000 / 500 ± 5 %
c	Order Quantity Tolerance		± 5%	± 5%
11.0	Marking on Outer Sheath			
a	By Printing		RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No	RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No
b	Sequential Length Marking		Shall be provided at every metere by printing	Shall be provided at every metere by printing

Note: 5 % of Ordered quantity may be supplied in Assorted Non-Standard length but shall not be less than 50 Metres

Note: In case of any dispute/typographical error interpretation of the IS specification shall prevail.

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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 6 SQMM	3C X 4 SQMM
1.0	General			
a	Brand Name	---	RALLISON	RALLISON
b	Standards applicable	---	IS: 17505 (Part 1) :2021 with up to date amendment	IS: 17505 (Part 1) :2021 with up to date amendment
c	Voltage Grade	Volts	1100 Volts	1100 Volts
d	Permissible voltage & frequency variation	%	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$	Voltage variation $\pm 10\%$, Frequency variation $\pm 5\%$
e	Suitable for earthed / unearthed systems		Suitable for earth and un earthed system	Suitable for earth and un earthed system
2.0	Current rating			
a	Continuous current rating under normal conditions			
i	In Ground at 30°C	Amps	56	45
ii	In Air at 40°C	Amps	52	41
b	Maximum conductor temperature for normal operation	°C	90	90
c	Maximum conductor temperature for short circuit rating	°C	250	250
d	Short circuit rating for 1 sec. duration	K.Amps	0.8586	0.5724
3.0	Conductor	---		
a	Material	---	Annealed Plain Copper	Annealed Plain Copper
b	Standard applicable	---	as per IS : 8130-2013 , Class-2	as per IS : 8130-2013 , Class-2
c	Shape of Conductor	---	Stranded, Circular	Stranded, Circular
d	No. of strands	Nos.	7	7
e	Dia of Strands (Before stranding)	mm	Diameter of strands shall be so choosen to meet the conductor Resistance	Diameter of strands shall be so choosen to meet the conductor Resistance
f	Nominal Cross Sectional Area Phase	sqmm	6	4
g	Nominal Cross Sectional Area Neutral	sqmm	NA	NA
h	Max. Conductor Resistance Phase at 20°C(DC)	Ω /KM	3.08	4.61
i	Max. Conductor Resistance Neutral at 20°C(DC)	Ω /KM	NA	NA
j	Max. Conductor Resistance Phase at 90°C	Ω /KM	3.9424	5.9008
k	Approximate Reactance of cable at 50 Hz for ref only	Ω /KM	0.085	0.087
l	Approximate Capacitance at 50Hz for ref. only	μ F/Km	0.13	0.11
4.0	Fire Barrier Tape		Glass Mica Tape (Fire Barrier Tape) 2 Tape shall be applied by helical wrapping with 100 % Coverage ,Over conductor throught the length .	

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TECHNICAL DATA SHEET FIRE SURVIVAL COPPER ARMoured CABLE

CUSTOMER :

PROJECT :

Sr. No.	Description	Unit	3C X 6 SQMM	3C X 4 SQMM
5.0	Insulation	---		
a	Material	---	XLPE	XLPE
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Nominal Thickness (Phase)	mm	0.7	0.7
d	Nominal Thickness (Neutral)		N/A	N/A
e	Minimum volume resistivity at 27°C	Ω-cm	1X10 ¹⁴	1X10 ¹⁴
f	Core identification		Red, Yellow & Blue	Red, Yellow & Blue
6.0	Inner Sheath	---		
a	Material	---	Thermoplastic material (PVC with FR Properties)	Thermoplastic material (PVC with FR Properties)
b	Nominal Thickness	mm	0.8	0.8
c	Minimum Thickness	mm	0.44	0.44
d	Extruded/ Wrapped		Extruded	Extruded
e	Colour		Black	Black
7.0	Armouring	---		
a	Material	---	Galvanized Steel	Galvanized Steel
b	Standard applicable	---	IS:3975/1999	IS:3975/1999
c	Type of Armour		Round Wire	Round Wire
d	Size of armour	mm	1.40 ± 0.040	0.90 ± 0.030
e	Binder Tape Over Armour		Synthetic tape may be applied by helical wrapping with 100 % coverage over armour	
8.0	Outer Sheath	---		
a	Material	---	LSZH	LSZH
b	Standard applicable	---	IS: 17505 (Part 1) :2021	IS: 17505 (Part 1) :2021
c	Minimum Thickness	mm	1.24	1.24
d	Nominal Thickness	mm	1.8	1.8
e	Overall diameter	mm	17	15
f	Tolerance in overall diameter	± mm	± 4	± 4
g	Outer Sheath colour		Black	Black
8.1	Circuit Integrity Test		Shall pass the test Categories C as per IS:17505 (Part-1) :2021 at 950 ±40 0 C for 3 Hours Category F	
9.0	Min. Bending Radius		12 X Cable Dia	12 X Cable Dia
10.0	Packing Detail			
a	Type of Packing Drum		Wooden Drum	Wooden Drum
b	Packing Length with Tolerance	metres	1000/500 ± 5%	1000 / 500 ± 5 %
c	Order Quantity Tolerance		± 5%	± 5%
11.0	Marking on Outer Sheath			
a	By Printing		RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No	RALLISON 1100V ELECTRIC SIZE OF CABLE ,CABLE CODE , YEAR OF MFG IS:17505 Part-1 ISI LOGO CML-0008600116113 BATCH No
b	Sequential Length Marking		Shall be provided at every metere by printing	Shall be provided at every metere by printing

Note: 5 % of Ordered quantity may be supplied in Assorted Non-Standard length but shall not be less than 50 Metres

Note: In case of any dispute/typographical error interpretation of the IS specification shall prevail.